Piston Driven Shock Waves in Non-Homogeneous Media

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1 Abstract

1. The problem settings that the article deals with - self-similar solutions of the power-law pressure driven shock waves in non-homogeneous media.

2. The main results are:

   (a) Showing that the solutions take various forms depending on the value of pressure & density exponents.

   (b) Analysis of the **shock propagation** - shock propagating at different accelarations (decelerating, constant velocity, accelerating).

   (c) Analysis of the **density near the piston** - either finite, vanishing or diverging..

   (d) Comparison of the results to **Lagrangian hydro simulations** - showing good agreement.

2 subsection

Introduction

1. Talking about the key role that analytic solutions of shock waves play in understanding phenomena in astrophysical & laboratory HEDP systems.

2. Mentions Shussman & Heizler (2015) and says that it provided a separate traetment of the ablation & shock region and thus enabled the use of **binary EOS**.

3. Gives a cool example of an experiment (ref. 11) where a the drive temperature in the hohlraum was measured according to the shock speed.

4. Mentions Shussman & Heizler again and says that they provided a solution for an homogeneous medium, and that the current work extends it to non-homogeneous media - solutions to ablation driven shock waves can be generalized after this paper. which is **useful for astrophysical models of the interior and atmosphere of stars & galaxies**.

5. Solutions are presented in both Lagrangian & Eulerian coordinates, and the transformation between them is given by a useful formula that **demonstrates the behavior of the shocked fluid in the x-t plane**.

6. The energy is also studied - via (1) its temporal dependence, and (2) its division into kinetic & internal energy.

7. The goal of the comparison to hydro simulations is to validate the simulations (surprising - I should ask Shay about it).

3 Statement of the problem

Problem settings:

1. The lagrangian coordinate is defined as:

$$m = \int_0^x \rho(x', t) dx' \tag{1}$$

2. The equations of motion are Euler equations (in Lag. coordinates) + the EOS (in Rosen's notation):

$$\frac{\partial V}{\partial t} - \frac{\partial u}{\partial m} = 0 \tag{2}$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \tag{3}$$

$$\frac{\partial e}{\partial t} + p\frac{\partial V}{\partial t} = 0 \tag{4}$$

$$pv = re \Leftrightarrow p = r\rho e \quad (r = \gamma - 1) \tag{5}$$

3. The boundary condition is a power-law pressure drive at the piston:

$$p(m=0,t) = p_0 t^{\tau} \tag{6}$$

4. The initial conditions are medium at rest with a power-law density profile:

$$u(m,t=0) = p(m,t=0) = 0 \tag{7}$$

$$\rho(m,t=0) = \rho_0 x^{-\omega} \Leftrightarrow v(x,t=0) = v_0 x^{\omega} \tag{8}$$

where the initial specific volume is $v_0 = \frac{1}{\rho_0}$.

5. One can integrate the density profile to get a relation of Lagrangian $\Leftrightarrow$ Eulerian coordinates:

$$m(x,t=0) = \int_0^x \rho_0(x')^{-\omega} dx' = \frac{\rho_0}{1-\omega} x^{1-\omega} \Leftrightarrow \boxed{x(m,t=0) = [v_0(1-\omega)m]^{\frac{1}{1-\omega}}} \tag{9}$$

which can be interpreted also as the initial specific volume in Lagrangian coordinates:

$$\boxed{v(m,t=0)} = v_0 x^{\omega} = \boxed{v_0^{\frac{1}{1-\omega}} \left[(1-\omega)m\right]^{\frac{\omega}{1-\omega}}} \tag{10}$$

4 Self-similar Representation

4.1 Using Pi theorem

The problem has 7 dimensional parameters:

1. Independent variables: t, m.
2. Dependent variables (hydrodynamic profiles): p, v, u.
3. Initial & boundary conditions: p_0, v_0

Therefore, there are $N = 7$ dimensional parameters, and $M = 3$ fundamental dimensions (M, L, T) - so we expect $N - M = 4$ dimensionless parameters, which will be chosen to be $\xi, V(\xi), U(\xi), P(\xi)$, where ξ is the self-similar space variable and V, U, P are the self-similar hydrodynamic profiles. The dimensional variables used to contruct the dimensionless parameters are v_0, p_0, t. If one solves the linear non-homogeneous equations for the

construction of ξ, v, u, p as power-laws of v_0, p_0, t, one gets:

$$\xi = m v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega-1}{2-\omega}} t^{\frac{(2+\tau)(\omega-1)}{2-\omega}} \tag{11}$$

$$v(m,t) = V(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(2+\tau)\omega}{2-\omega}} \tag{12}$$

$$u(m,t) = U(\xi) v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \tag{13}$$

$$p(m,t) = P(\xi) p_0 t^{\tau} \tag{14}$$

The Pi-table is given in the article (table 1).

4.2 Boundary conditions and self-similar ODEs

The boundary condition at the piston ($m = 0$) in terms of the dimensionless profiles is:

$$P(\xi = 0) = 1 \tag{15}$$

By using the relations:

$$\begin{aligned} \frac{\partial \xi}{\partial m} &= \frac{\xi}{m} \\ \frac{\partial \xi}{\partial t} &= \frac{(2+\tau)(\omega-1)}{2-\omega} \frac{\xi}{t} \end{aligned} \tag{16}$$

one achieved a set of 3 coupled ODEs for the self-similar profiles:

$$\left(\frac{2+\tau}{2-\omega}\right)\left((\omega-1)\xi V' + \omega V\right) - U' = 0 \tag{17}$$

$$\frac{(2+\tau)(\omega-1)}{2-\omega}\xi U' + U\left(\frac{\omega+\tau}{2-\omega}\right) + P' = 0 \tag{18}$$

$$\frac{(2+\tau)\omega}{2-\omega}\xi V' + V\left(\frac{(2+\tau)\omega}{2-\omega}\right) + P\left(\frac{(2+\tau)(\omega-1)}{2-\omega}\right) + r\left(\frac{(2+\tau)(\omega-1)}{2-\omega}\right)U = 0 \tag{19}$$

The last equation here should be cross checked. which the handy copilot arranged to solve for the derivatives vector:

$$\frac{dV}{d\xi} = \frac{V}{\xi}\frac{(2+\tau)(\omega-1)V + (2-\omega)U}{(1-\omega)V - U} \tag{20}$$

$$\frac{dU}{d\xi} = \frac{U}{\xi}\frac{(2+\tau)(\omega-1)V + (2-\omega)U + (1-\omega)P}{(1-\omega)V - U} \tag{21}$$

$$\frac{dP}{d\xi} = \frac{P}{\xi}\frac{(2+\tau)(\omega-1)V + (2-\omega)U + r(1-\omega)U}{(1-\omega)V - U} \tag{22}$$

It is mentioned that in case of $\omega = 0$ (homogeneous medium) the equations reduce to those of Shussman & Heizler (2015).

5 Jump conditions at the shock front

5.1 Reminder of the problem settings

1. In order to solve the ODEs, one should be reminded of the problem settings. In the unshocked region - the hydrodynamic profiles are given simply by the initial conditions (power-law density, zero velocity & pressure). In the shocked region - the hydrodynamic profiles are given by the self-similar solution to the ODEs.

2. The transition between the shocked & unshocked regions is the shock front, and the jump conditions at the shock front are given by the Hugoniot conditions.

5.2 Derivation of the Rankine-Hugoniot conditions

1. Claim: The general form of the Rankine-Hugoniot conditions for a set of conserved quantities $\boldsymbol{u}(y,t)$ which obey a set of (nonlinear) hyperbolic conservation laws in the form $\frac{\partial \boldsymbol{u}}{\partial t} + \frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial y} = 0$ is given by:

$$\dot{y}_s \Delta \boldsymbol{u} = \Delta \boldsymbol{f} \tag{23}$$

   where $\Delta \boldsymbol{u} = \boldsymbol{u}_2 - \boldsymbol{u}_1$ is the jump in the conserved quantities across the shock, and $\Delta \boldsymbol{f} = \boldsymbol{f}(\boldsymbol{u}_2) - \boldsymbol{f}(\boldsymbol{u}_1)$ is the jump in the fluxes across the shock. $\dot{y}_s$ is the local discontinuity velocity (e.g. shock velocity). The proof is not given in the article, but copilot gave it to me and it appears on the appendix of this document.

2. The article says that it needs to convert the energy conservation equation (eq. 4 in the article) to a conservation law form:

$$\frac{\partial}{\partial t}\left(\frac{pv}{r} + \frac{1}{2}u^2\right) + \frac{\partial pu}{\partial m} = 0 \tag{24}$$

   the derivation of this equation is also given in the appendix.

3. Using the latter equations along with the mass & momentum conservation equations, one can apply the generalized Rankine-Hugoniot

conditions to get the jump conditions at the shock front:

$$\dot{m}_s(v_s - v_{ref}) = u_{ref} - u_s \tag{25}$$

$$\dot{m}_s(u_s - u_{ref}) = p_s - p_{ref} \tag{26}$$

$$\dot{m}_s\left(\frac{p_s v_s}{r} + \frac{1}{2}u_s^2\right) = u_s p_s - u_{ref} p_{ref} \tag{27}$$

where 'ref' stands for the (referenced) unshocked region and 's' stands for the shocked region. $\dot{m}_s$ is the shock velocity in Lagrangian coordinates.

5.3 Simplification of the jump conditions

1. Using Initial Conditions: Since the unshocked region satisfies the initial conditions $u_{ref} = p_{ref} = 0$, the jump conditions can be simplified to (what is actually **the strong shock limit**):

$$\dot{m}_s v_s = -u_s \tag{28}$$

$$\dot{m}_s u_s = p_s \tag{29}$$

$$\dot{m}_s\left(\frac{p_s v_s}{r} + \frac{1}{2}u_s^2\right) = u_s p_s \tag{30}$$

2. Using the profile of $m(t)$ to find $\dot{m}_s(t)$: Through the inversion of the relation between ξ and m, one can find the profile of $m(t)$:

$$m(t) = v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{\frac{(2+\tau)(1-\omega)}{2-\omega}} \xi \tag{31}$$

so the shock velocity in Lagrangian coordinates is given by:

$$\dot{m}_s(t) = \frac{(2+\tau)(1-\omega)}{2-\omega} v_0^{-\frac{1}{2-\omega}} p_0^{\frac{1-\omega}{2-\omega}} t^{\frac{(2+\tau)(1-\omega)}{2-\omega}-1} \xi_s \tag{32}$$

where ξ_s is the self-similar coordinate of the shock front **which is NOT KNOWN A-PROIRI and should be found by using a shooting method to solve the ODEs** - described in the next section. NOTE: Plotting $\frac{(2+\tau)(1-\omega)}{2-\omega}$ as a function of ω, τ (Fig. 2 in the article) shows that the requirement for the shock to propagate is $\tau > -2$ (remember that $\omega < 1$ for the mass to be finite, and **the sign of $\frac{(2+\tau)(1-\omega)}{2-\omega}$ determines the direction of the shock propagation**).

3. substituting Self-similar profiles: Notice that v_{ref} is also known in terms of the self-similar profiles (we calculated the initial specific volume in Lagrangian coordinates in the previous section):

$$\begin{aligned} v_{ref} &= v(m, t=0) = v_0^{\frac{1}{1-\omega}} \left[(1-\omega)m\right]^{\frac{\omega}{1-\omega}} \\ v_{ref} &= (1-\omega)^{\frac{\omega}{1-\omega}} v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(2+\tau)\omega}{2-\omega}} \xi^{\frac{\omega}{1-\omega}} \end{aligned} \tag{33}$$

writing the hydrodynamic profiles in terms of the self-similar profiles evaluated at the shock front, we get:

$$\begin{aligned} v_s &= V_s v_0^{\frac{1}{2-\omega}} p_0^{\frac{\omega}{2-\omega}} t^{\frac{(2+\tau)\omega}{2-\omega}} \\ u_s &= U_s v_0^{\frac{1}{2-\omega}} p_0^{\frac{1}{2-\omega}} t^{\frac{\omega+\tau}{2-\omega}} \\ p_s &= P_s p_0 t^{\tau} \end{aligned} \tag{34}$$

where V_s, U_s, P_s are the self-similar profiles evaluated at the shock front ξ_s. Substituting these into the jump conditions and taking advantage of the **vanishing of dimensional parameters** we get the dimensionless jump conditions:

$$\begin{aligned} V_s &= ((1-\omega)\xi_s)^{\frac{\omega}{1-\omega}} - \frac{U_s}{(2+\tau)\left(\frac{1-\omega}{2-\omega}\right)\xi_s} \\ P_s &= (2+\tau)\left(\frac{1-\omega}{2-\omega}\right)\xi_s U_s \\ (2+\tau)\left(\frac{1-\omega}{2-\omega}\right)\left(\frac{P_s V_s}{r} + \frac{1}{2}U_s^2\right) &= U_s P_s \end{aligned} \tag{35}$$

which can be simplified to:

$$\begin{aligned} V_s &= \frac{r}{r+2}\left((1-\omega)\xi_s\right)^{\frac{\omega}{1-\omega}} \\ U_s &= \frac{2}{r}(2+\tau)\left(\frac{1-\omega}{2-\omega}\right)\xi_s U_s \\ P_s &= \frac{U_s^2}{2}\frac{r}{V_s} \end{aligned} \tag{36}$$

As before, substituting $\omega = 0$ (homogeneous medium) reduces the jump conditions to those of Shussman & Heizler (2015).

6 Appendix

6.1 Proof of generalized Rankine-Hugoniot Jump Conditions

1. Proof: Consider the integral form of the conservation law:

$$\frac{d}{dt}\int_{y_1}^{y_2} \boldsymbol{u}(y,t)dy + \boldsymbol{f}(\boldsymbol{u}(y_2,t)) - \boldsymbol{f}(\boldsymbol{u}(y_1,t)) = 0 \tag{37}$$

where $y_1 < y_s(t) < y_2$ are two fixed points on the left & right of the shock front. The integral can be split into two parts:

$$\int_{y_1}^{y_2} \boldsymbol{u}(y,t)dy = \int_{y_1}^{y_s(t)} \boldsymbol{u}(y,t)dy + \int_{y_s(t)}^{y_2} \boldsymbol{u}(y,t)dy \tag{38}$$

Taking the time derivative of the integral, we get:

$$\frac{d}{dt}\int_{y_1}^{y_2} \boldsymbol{u}(y,t)dy = \int_{y_1}^{y_s(t)} \frac{\partial \boldsymbol{u}}{\partial t}dy + \int_{y_s(t)}^{y_2} \frac{\partial \boldsymbol{u}}{\partial t}dy + \dot{y}_s(\boldsymbol{u}_2 - \boldsymbol{u}_1) \tag{39}$$

where we used the Leibniz rule for differentiation under the integral sign. Substituting the conservation law into the integrals, we get:

$$\frac{d}{dt}\int_{y_1}^{y_2} \boldsymbol{u}(y,t)dy = -\int_{y_1}^{y_s(t)} \frac{\partial \boldsymbol{f}}{\partial y}dy - \int_{y_s(t)}^{y_2} \frac{\partial \boldsymbol{f}}{\partial y}dy + \dot{y}_s(\boldsymbol{u}_2 - \boldsymbol{u}_1) \tag{40}$$

Evaluating the integrals, we get:

$$\frac{d}{dt}\int_{y_1}^{y_2} \boldsymbol{u}(y,t)dy = -\boldsymbol{f}(\boldsymbol{u}_2) + \boldsymbol{f}(\boldsymbol{u}_1) + \dot{y}_s(\boldsymbol{u}_2 - \boldsymbol{u}_1) \tag{41}$$

Substituting this back into the integral form of the conservation law, we get:

$$-\boldsymbol{f}(\boldsymbol{u}_2) + \boldsymbol{f}(\boldsymbol{u}_1) + \dot{y}_s(\boldsymbol{u}_2 - \boldsymbol{u}_1) + \boldsymbol{f}(\boldsymbol{u}_2) - \boldsymbol{f}(\boldsymbol{u}_1) = 0 \tag{42}$$

which simplifies to the Rankine-Hugoniot condition:

$$\dot{y}_s(\boldsymbol{u}_2 - \boldsymbol{u}_1) = \boldsymbol{f}(\boldsymbol{u}_2) - \boldsymbol{f}(\boldsymbol{u}_1) \tag{43}$$

2. Proof of the conversion of the energy conservation equation to a conservation law form: Starting from the energy conservation equation in Lagrangian coordinates:

$$\frac{\partial e}{\partial t} + p\frac{\partial V}{\partial t} = 0 \tag{44}$$

we can use the EOS to express e in terms of p and v:

$$e = \frac{pv}{r} \tag{45}$$

Substituting this into the energy conservation equation, we get:

$$\frac{\partial}{\partial t}\left(\frac{pv}{r}\right) + p\frac{\partial V}{\partial t} = 0 \tag{46}$$

which under the assumption of an ideal gas can be rewritten as:

$$\frac{\partial}{\partial t}\left(\frac{pv}{r} + \frac{1}{2}u^2\right) + \frac{\partial pu}{\partial m} = 0 \tag{47}$$

where the kinetic energy term $\frac{1}{2}u^2$ is added to the internal energy term $\frac{pv}{r}$ to form the total energy, and the flux term $\frac{\partial pu}{\partial m}$ represents the work done by pressure forces. This equation is now in the form of a conservation law, which allows us to apply the Rankine-Hugoniot conditions at the shock front. I actually need to make sure that it makes sense to add the kinetic energy and transform the work term as the copilot did. Need to make sure it makes sense with Zeldovich.

7 Balak presentation

$$\frac{1}{\nu}\frac{\partial \psi}{\partial t} + \mu\frac{\partial \psi}{\partial x} + \sigma_t\psi = \frac{c}{2}\sigma_t\int_{-1}^{1}\psi(\mu')d\mu' + Q \tag{48}$$

after integegration over μ and over $\mu d\mu$ one gets:

$$\frac{1}{\nu}\frac{\partial \phi}{\partial t} \tag{49}$$